

Roll No.

TBT-301

B. TECH. (BIOTECHNOLOGY)

(THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

GENETICS AND MOLECULAR BIOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What are traits and alleles ? Describe the possible traits can be passed down to offspring. (CO1)

OR

- (b) Describe the Mendel's law which form the basis of genetics. Give example. (CO1)

2. (a) Explain epigenetics with *two* examples. (CO1)

OR

- (b) What are chromosomal disorders ? How are they different from infectious diseases ? Give examples. (CO1)

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(2)

3. (a) What is sex-linked inheritance ? How is inheritance of different disorders linked to X-chromosomes ? (CO1/CO2)

OR

- (b) Describe how RNA was proved as a genetic material. (CO1/CO2)

4. (a) How was DNA discovered as genetic material ? Explain. (CO2)

OR

- (b) How the transforming principle was proven by Hershey and Chase as DNA ? (CO2)

5. (a) Describe the chemical structure of DNA double helical model with all bonding. (CO2)

OR

- (b) Describe the differences in the organization of prokaryotic and eukaryotic genome. (CO2)